

CLIENT DELIVERABLE

Systack — Invoice Parser

Systack Documentation

ROLE	COLOR	HEX
Headers, CTAs	Navy	 #001a2d
Navy Light	Navy Light	 #002845
Secondary accents	Teal	 #007da9
Primary buttons, links	Cyan	 #00a1db
Gradients	Cyan Bright	 #00c5e0
Backgrounds	Gray 50	 #f8fafc
Cards	Gray 100	 #f1f5f9
Borders	Gray 200	 #e2e8f0
Muted text	Gray 400	 #94a3b8
Body text	Gray 600	 #475569
Headings	Gray 800	 #1e293b
Success	Green	 #22c55e
Error	Red	 #ef4444
Accent highlights	Purple	 #8b5cf6

Systack — Invoice Parser

Vendor Format Guide

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Version: 1.0

Status: Internal — Systack Only

Source System: Systack Invoice Parser API (Live since 2026-06-08)

Date: 2026-06-16

Builder: SOL / Assembly

Brand Reference

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1. Overview

The Invoice Parser uses format-specific handlers to extract structured data from known vendor invoice layouts. When a vendor format is not recognized, the system falls back to OCR.

This guide explains how to add support for new vendor formats.

2. How Format Detection Works

```
PDF received
  → Extract text content
    → Match against known vendor signatures
      → Found? → Use format-specific handler
      → Not found? → Use OCR fallback
```

Vendor signatures are unique text patterns that identify a vendor:

- Company name in header
- Specific formatting patterns
- Unique field labels (e.g., “AT&T Account Number”)
- Logo text recognition

3. Currently Supported Vendors

VENDOR	HANDLER	SIGNATURE PATTERN
AT&T	<code>att_handler</code>	“AT&T” + account number format
Verizon	<code>verizon_handler</code>	“Verizon” + billing period
Utility (generic)	<code>utility_handler</code>	“Service Address” + meter number
Wholesale (generic)	<code>wholesale_handler</code>	“PO Number” + quantity columns
Service (generic)	<code>generic_handler</code>	“Invoice Date” + “Service Description”

4. Adding a New Vendor Format

Step 1 — Collect Sample Invoices

Gather 3–5 sample PDF invoices from the vendor. Variety helps:

- Different billing periods
- Different amounts
- Different line item counts

Step 2 — Identify Extraction Fields

Map the vendor’s invoice layout to our standard fields:

OUR FIELD	VENDOR’S LABEL	LOCATION ON PAGE
<code>vendor</code>	Company name	Top header
<code>invoice_number</code>	“Invoice #”, “Bill #”, “Reference”	Top right or below header
<code>date</code>	“Invoice Date”, “Bill Date”	Near invoice number
<code>line_items[].name</code>	Item description column	Table body
<code>line_items[].qty</code>	“Qty”, “Units”	Table column
<code>line_items[].price</code>	“Rate”, “Unit Price”	Table column
<code>line_items[].total</code>	“Amount”, “Charge”	Table column
<code>subtotal</code>	“Subtotal”, “Net Amount”	Below table
<code>tax</code>	“Tax”, “GST”, “VAT”	Below subtotal
<code>total</code>	“Total”, “Amount Due”, “Balance”	Bottom of invoice

Step 3 — Write the Handler

Create a new handler in the parser codebase:

```
# handlers/vendor_name_handler.py

def extract(pdf_text: str) -> dict:
```

```

"""
Extract invoice data from Vendor Name PDF format.

Args:
    pdf_text: Raw text extracted from PDF

Returns:
    Dict with standard invoice fields
"""

result = {
    "vendor": "Vendor Name",
    "invoice_number": None,
    "date": None,
    "line_items": [],
    "subtotal": None,
    "tax": None,
    "total": None,
}

# 1. Extract invoice number
# Look for pattern: "Invoice #: XXXXX"
import re
inv_match = re.search(r'Invoice\s*#:\s*(\S+)', pdf_text)
if inv_match:
    result["invoice_number"] = inv_match.group(1)

# 2. Extract date
date_match = re.search(r'Date:\s*(\d{1,2}/\d{1,2}/\d{4})', pdf_text)
if date_match:
    result["date"] = date_match.group(1)

# 3. Extract line items from table
# ... vendor-specific table parsing logic ...

# 4. Extract totals
total_match = re.search(r'Total\s*\$?([\d,]+\.\d{2})', pdf_text)
if total_match:
    result["total"] = float(total_match.group(1).replace(',', ''))

return result

```

Step 4 — Register the Handler

Add to the handler registry:

```

# handlers/registry.py

HANDLERS = {
    "att": att_handler.extract,
    "verizon": verizon_handler.extract,
    "vendor_name": vendor_name_handler.extract, # NEW
}

```

Add signature detection:

```

# handlers/detector.py

SIGNATURES = {
    "AT&T": "att",

```

```
"Verizon": "verizon",
"Vendor Name": "vendor_name", # NEW
}
```

Step 5 — Test

```
# Test with sample invoice
curl -X POST https://invoices.systack.net/extract \
  -F "file=@sample_vendor_invoice.pdf"

# Verify response
{
  "vendor": "Vendor Name",
  "invoice_number": "12345",
  ...
}
```

Step 6 — Document

Add to this guide:

- Vendor name and handler file
- Signature pattern used
- Any special extraction logic
- Known limitations

5. Handler Design Guidelines

Do

- Use regex patterns that are specific enough to avoid false matches
- Handle variations in formatting (spacing, punctuation)
- Return `None` for fields that can't be confidently extracted
- Test with multiple sample invoices
- Handle multi-page invoices (check all pages)

Don't

- Hardcode dollar amounts or dates
- Assume table column positions (use labels, not positions)
- Crash on unexpected formats (return partial data gracefully)
- Use filename for vendor detection (use content)

6. OCR Fallback

When no format handler matches:

1. PDF is rendered to image

2. Tesseract OCR extracts all visible text
3. Generic patterns are applied:
 - Largest dollar amount → likely total
 - Date-like patterns → likely invoice date
 - Number with 4+ digits → likely invoice number
 - Company name at top → likely vendor

OCR limitations:

- Handwriting not recognized
- Complex table layouts may scramble
- Confidence varies by PDF quality

7. Testing New Formats

Test Script

```
#!/bin/bash
# test_vendor_format.sh

VENDOR_DIR="./test_invoices/vendor_name"

for pdf in "$VENDOR_DIR"/*.pdf; do
    echo "Testing: $pdf"
    curl -s -X POST https://invoices.systack.net/extract \
        -F "file=@$pdf" | python3 -m json.tool
    echo "----"
done
```

Validation Checklist

- ☐ Vendor name correctly identified
- ☐ Invoice number extracted
- ☐ Date parsed correctly
- ☐ All line items captured
- ☐ Quantities match invoice
- ☐ Prices match invoice
- ☐ Subtotal calculated correctly
- ☐ Tax extracted (if present)
- ☐ Total matches invoice
- ☐ Works across 3+ sample invoices

8. Version History

VERSION	DATE	CHANGE
1.0	2026-06-08	Initial 9 handlers deployed

VERSION	DATE	CHANGE
1.0	2026-06-16	Documented (this file)

Internal document — do not share with clients.
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